Gunda Sai Yashaswith

es24btech11015@iith.ac.in | saiyashaswithgunda@gmail.com | +91 9121318572 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.31
Sri Chaitanya Junior College	Class XII (Telangana Board Of Intermediate Education(TGBIE))	2024	98.30%
Sri Chaitanya School	Class X (Board Of Secondary Education ,Telangana(BSE Telangana))	2022	10.00%

SCHOLASTIC ACHIEVEMENTS

Received the **Academic Excellence Award** for securing the **highest GPA** in the Engineering Science branch for **Calendar Year 2025 (Semesters 2 & 3).**

SKILLS

- **Languages:** C++, Python, SQL, C, Java
- **Libraries and Frameworks :** PyTorch, NumPy, Matplotlib, Pandas
- **Core Concepts :** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems
- **Developer Tools:** Git, GitHub, VS Code, CMake

PROJECTS

Persistent Key-Value Storage Engine | C++17 | GitHub

LevelDB-inspired persistent key-value storage engine built from scratch in C++, implementing core database internals without any external libraries.

- Implemented a page-based storage engine (4 KB fixed-size pages) with a **custom hash index (separate chaining)** mapping keys to page IDs for expected O(1) lookups, and an **interactive CLI** for real-time operations and benchmarking.
- Implemented an O(1) **LRU buffer pool** using a doubly linked list and hash map, with **deferred dirty-page writeback** (no-force policy) to minimize disk I/O; achieved ~**500K GET ops/sec** (cache hit) and **~100K ops/sec** (cache miss).
- Implemented **Write-Ahead Logging (WAL)** with **redo-based** crash recovery and single-pass log replay on restart , guaranteeing **durability** (ACID-D).
- Implemented **log compaction** for dead-page reclamation and periodic **checkpointing** every 100 operations, eliminating per-operation full index rewrites and **bounding WAL growth**; improved **SET** throughput by **~20x (220 to 4500 ops/sec).**

GPT-2 Implementation & Fine-Tuning Framework | Python, PyTorch | GitHub

Implemented the GPT-2 architecture from scratch with support for pre-trained weight loading, fine-tuning, and

autoregressive text generation.

- Built a decoder-only Transformer comprising **token and positional embeddings**, masked **multi-head self-attention**, **layer normalization, residual connections**, **GELU-based feed-forward networks**, and an **output projection** for autoregressive next-token prediction.
- Developed a **training and inference pipeline** with BPE tokenization, data preprocessing, **checkpoint loading**, and support for official GPT-2 (124M/355M) pre-trained weights.
- Loaded official OpenAI GPT-2 pre-trained weights into the custom implementation and **fine-tuned** the model for **SMS spam classification** (**95%** validation accuracy, **<0.30** validation loss) and **Alpaca-formatted instruction-following tasks**.
- Developed a FastAPI and a Streamlit web interface for interactive text generation and spam classification via a web interface.

BigInt Library | C++

- Developed a **custom arbitrary-precision integer library** using dynamic digit-vector representation, supporting signed integers of unlimited size beyond native data type limits.
- Implemented overloaded arithmetic (+, -, *, /, %) and comparison operators with carry/borrow management, long multiplication, long division, input validation, and exception handling.

RELEVANT COURSES

- **Core Courses:** Introduction to Programming, Data Structures & Applications, Algorithms, Database Management Systems , Operating Systems, Introduction to Computing , Theory of Computation, Discrete Mathematics for Computer Science
- **Math Courses :** Calculus, Probability & Random Variables, Introduction to Statistics, Differential Equations, Complex Variables, Elementary Linear Algebra

CODING PROFILES

- **LeetCode:** **sai_yashaswith** - Solved **400+** coding problems with an **ongoing 125+ day** problem-solving streak.